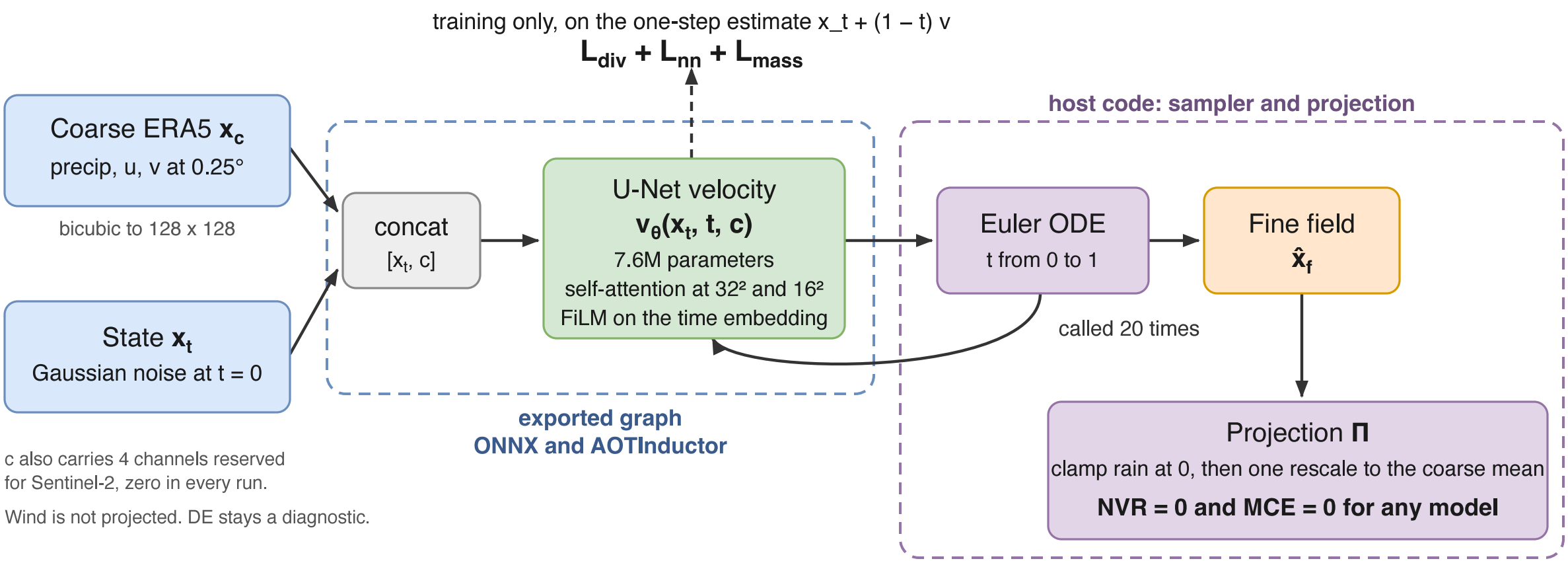

training only, on the one-step estimate x_t + (1 – t) v
$L_{div} + L_{nn} + L_{mass}$
Coarse ERA5 x_c
precip, u, v at 0.25°
bicubic to 128 x 128
State x_t
Gaussian noise at t = 0
c also carries 4 channels reserved for Sentinel-2, zero in every run.
Wind is not projected. DE stays a diagnostic.
concat
$[x_t, c]$
U-Net velocity
$v_\theta(x_t, t, c)$
7.6M parameters
self-attention at 32^2 and 16^2
FiLM on the time embedding
exported graph
ONNX and AOTInductor
host code: sampler and projection
Euler ODE
t from 0 to 1
called 20 times
Fine field
$\hat{x}_f$
Projection Π
clamp rain at 0, then one rescale to the coarse mean
NVR = 0 and MCE = 0 for any model